JAYAKANTH ELUMALAI

M.Eng. Student | Quality Engineering | Test & Measurement | Sensor Systems

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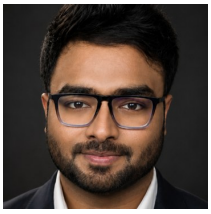

PROFILE

M.Eng. student in Analytical Instruments, Measurement and Sensor Technology at Coburg University of Applied Sciences with two years of industrial experience in quality assurance, testing and technical data analysis of electronic products (refrigerators). Experienced in evaluating measurement data, root-cause analysis and ISO/IEC 17025 laboratory documentation. **Seeking a Werkstudent position or internship (Pflichtpraktikum, to be credited toward my study programme)** in quality engineering, test & validation, measurement technology or sensor systems.

WORK EXPERIENCE

Quality Engineer – Testing & Data Analysis Jun 2023 – Sep 2025
Samsung India Electronics Pvt Ltd – Graduate Engineer Trainee (Jun 2023 – May 2024), then Associate Engineer (Jun 2024 – Sep 2025)

- Ran performance, durability and reliability tests on refrigerator products: temperature performance and distribution, pull-down time, energy consumption, defrost cycle, door/hinge durability cycling, gasket/seal leakage, noise and vibration, climate chamber (hot/humid), transport/drop/packaging, electrical safety and EMC testing.
- Analyzed test and measurement data (data loggers, thermocouples, DAQ; advanced Excel, Minitab, statistics, MATLAB/Simulink, Python) to identify deviations, trends and process abnormalities.
- Supported defect investigations and root-cause analysis; analyzed recurring quality issues to prioritize investigations and support corrective actions.
- Maintained test records, laboratory reports, SOPs and measurement traceability records in an ISO/IEC 17025 environment.
- Improved SOPs and testing procedures, contributing to a **20% increase in testing efficiency**.
- Worked with cross-functional teams, communicated technical findings and prepared work instructions and technical documentation for knowledge transfer.

EDUCATION

M.Eng. – Analytical Instruments, Measurement and Sensor Technology 2026 – present
Coburg University of Applied Sciences, Germany – incl. exchange semester at the University of Shanghai for Science and Technology (USST), Mar – Jul 2026 | German grade: 1.6

B.Eng. – Electrical and Electronics Engineering 2020 – 2023
Rajalakshmi Engineering College, Chennai, India | German grade: 2.1

Diploma – Electrical and Electronics Engineering (DEEE) 2017 – 2020
T.S. Srinivasan Center for Polytechnic College, Chennai, India | German grade: 2.0

SKILLS

Quality & testing: Quality assurance, reliability testing, root-cause analysis, process improvement, MSA, test automation / ATE

Measurement & data: Measurement data analysis, sensor technology, vibration analysis, technical reporting & documentation

Standards & methods: ISO/IEC 17025 procedures, calibration, EMC/EMV testing, SOPs & traceability

Tools: MATLAB / Simulink / Xcos, Python, MS Excel, Minitab, PLC, data loggers / DAQ

PROJECTS

- Intelligent Sensor Fault Detection System (MATLAB / Xcos)
- Autonomous Emergency Braking System (MATLAB / Xcos)
- IoT-Based Health and Safety Monitoring System for EV Batteries
- Arduino-based benerlon wheel
- Power Quality Improvement in a Three-Phase Voltage Source Converter Using Digital Communication Modulation for Non-Linear Load

Certifications

- Internal Auditor – Laboratory Management System (ISO 17025:2017), TÜV SÜD South Asia
- ServiceNow Certified System Administrator

Languages

- German – A2 (working toward B1)
- English – B2
- Tamil – fluent